

Acceptance Discrepancy Investigation

Comparing E906 Acceptance Calculations

Chatura Kuruppu

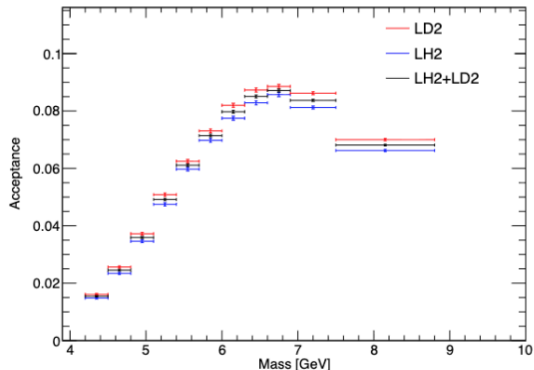
Fermilab / NMSU

September 3, 2026

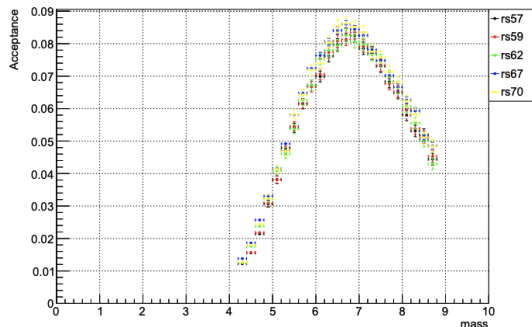
Initial Discrepancy: Overall Acceptance

Chatura's Acceptance

Integrated Acceptance vs mass



Kenichi's Acceptance

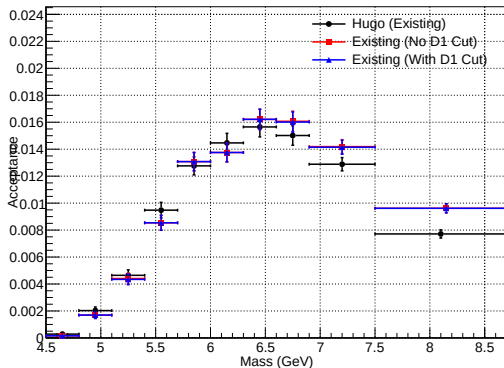


- Comparing the newly calculated acceptance to Kenichi's recent ROOT files.
- Next slides breakdown the discrepancy by x_F bins (Hugo vs. No D1 Cut vs. With D1

Acceptance Comparison: x_F Bin 0

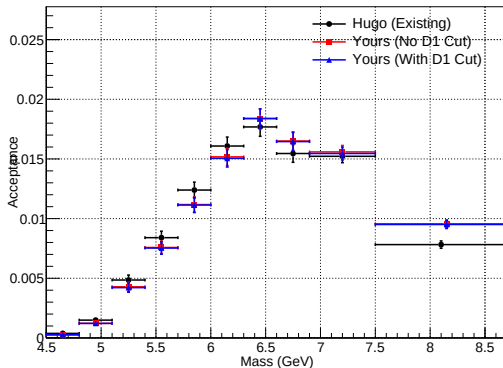
LH2 Target

LH2 Acceptance: x_F Bin 0



LD2 Target

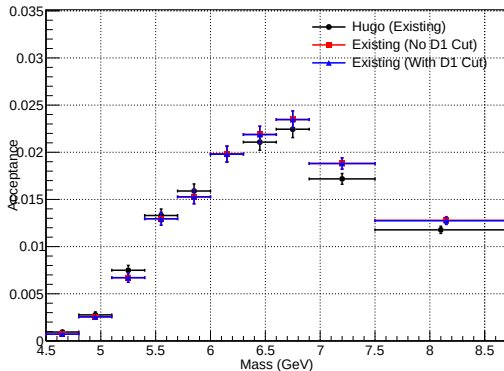
LD2 Acceptance: x_F Bin 0



Acceptance Comparison: x_F Bin 1

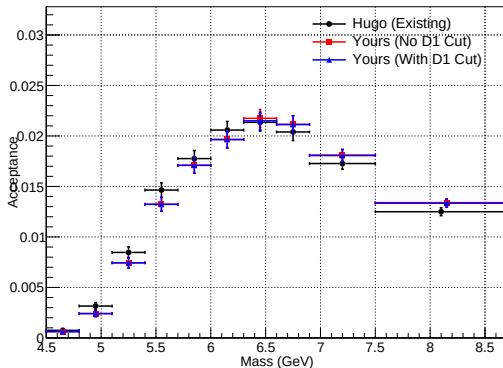
LH2 Target

LH2 Acceptance: x_F Bin 1



LD2 Target

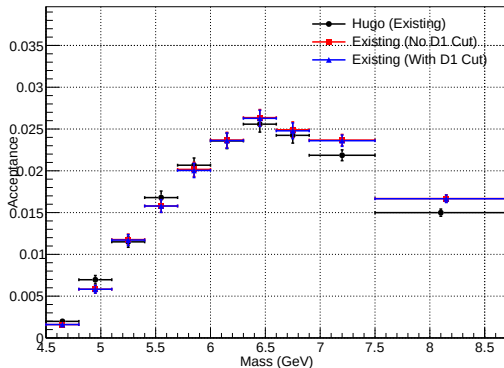
LD2 Acceptance: x_F Bin 1



Acceptance Comparison: x_F Bin 2

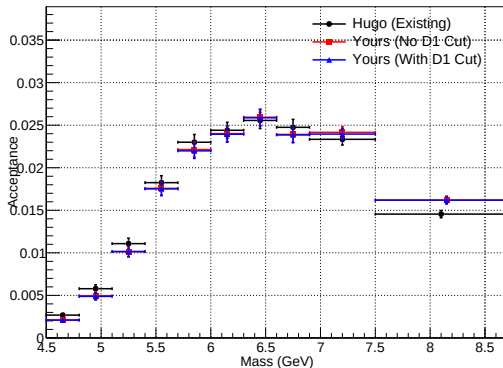
LH2 Target

LH2 Acceptance: x_F Bin 2



LD2 Target

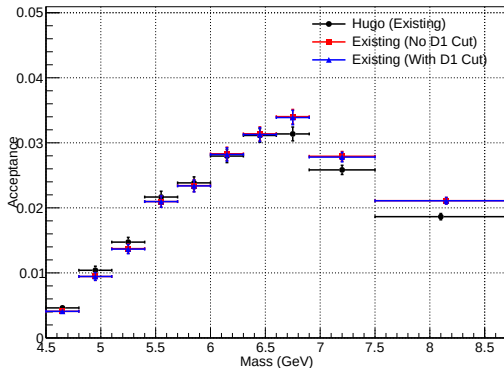
LD2 Acceptance: x_F Bin 2



Acceptance Comparison: x_F Bin 3

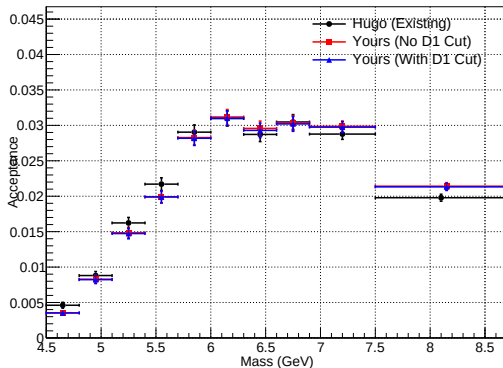
LH2 Target

LH2 Acceptance: x_F Bin 3



LD2 Target

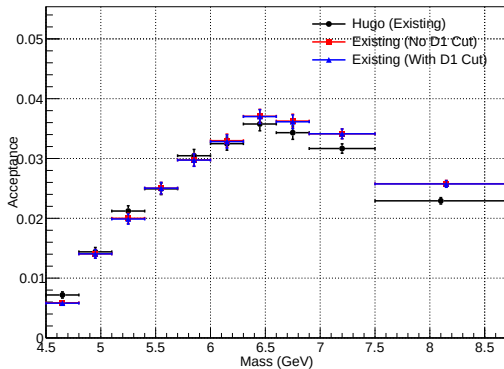
LD2 Acceptance: x_F Bin 3



Acceptance Comparison: x_F Bin 4

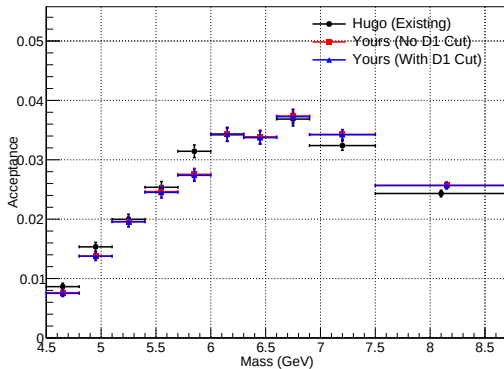
LH2 Target

LH2 Acceptance: x_F Bin 4



LD2 Target

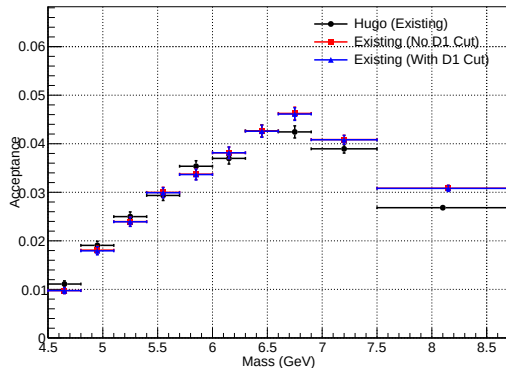
LD2 Acceptance: x_F Bin 4



Acceptance Comparison: x_F Bin 5

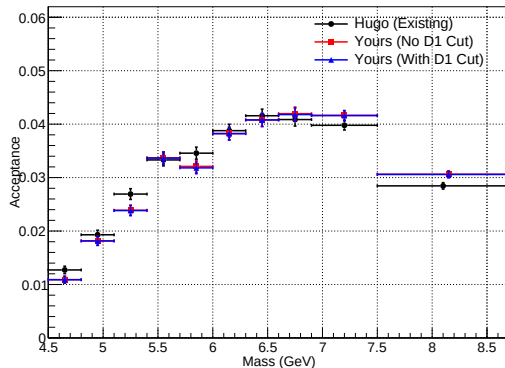
LH2 Target

LH2 Acceptance: x_F Bin 5



LD2 Target

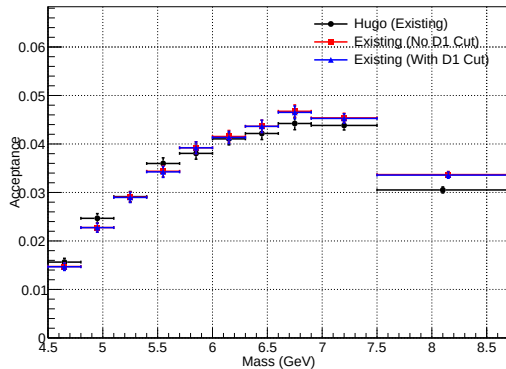
LD2 Acceptance: x_F Bin 5



Acceptance Comparison: x_F Bin 6

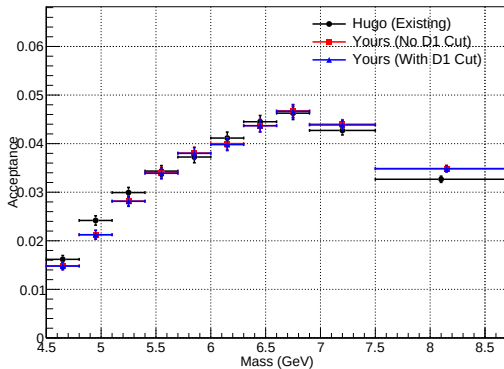
LH2 Target

LH2 Acceptance: x_F Bin 6



LD2 Target

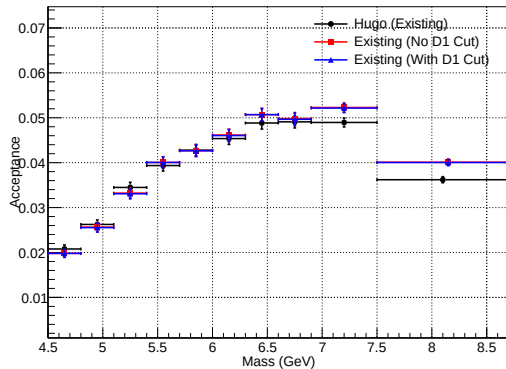
LD2 Acceptance: x_F Bin 6



Acceptance Comparison: x_F Bin 7

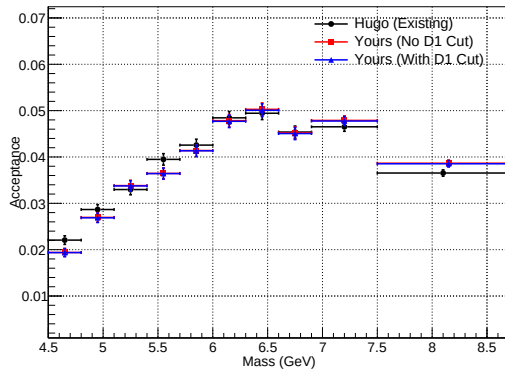
LH2 Target

LH2 Acceptance: x_F Bin 7



LD2 Target

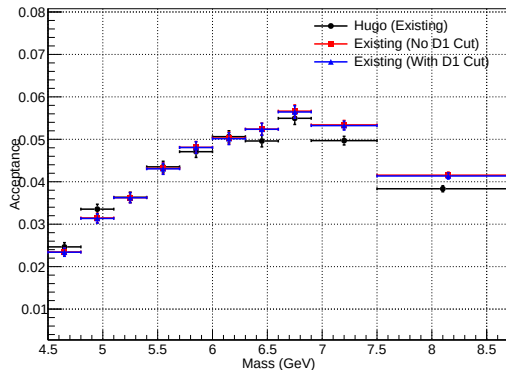
LD2 Acceptance: x_F Bin 7



Acceptance Comparison: x_F Bin 8

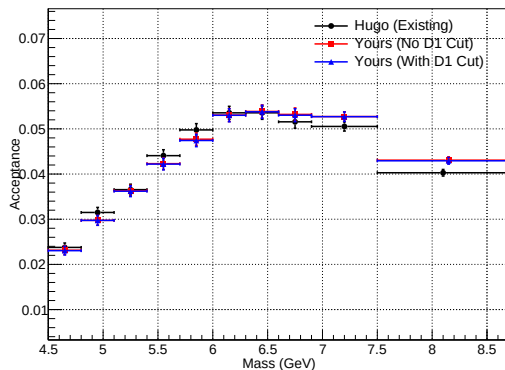
LH2 Target

LH2 Acceptance: x_F Bin 8



LD2 Target

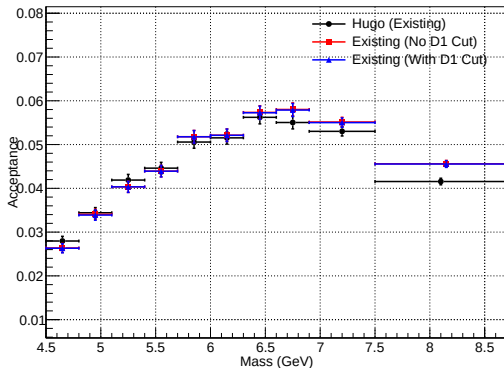
LD2 Acceptance: x_F Bin 8



Acceptance Comparison: x_F Bin 9

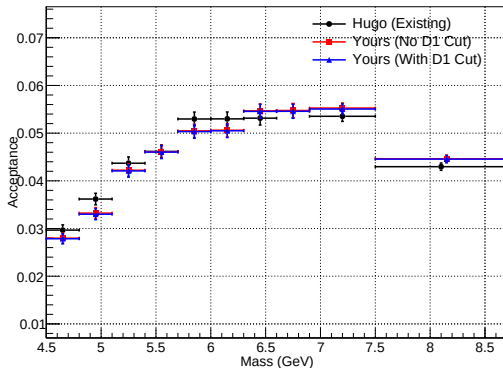
LH2 Target

LH2 Acceptance: x_F Bin 9



LD2 Target

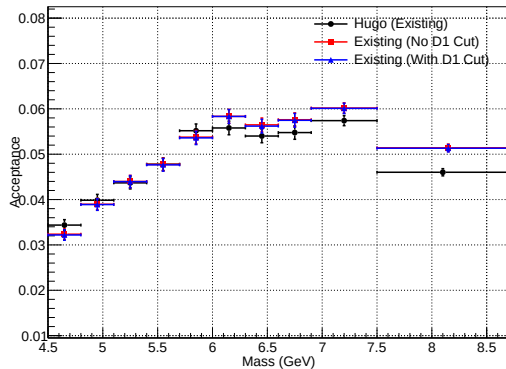
LD2 Acceptance: x_F Bin 9



Acceptance Comparison: x_F Bin 10

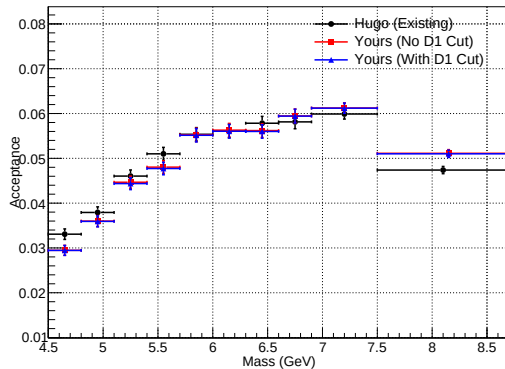
LH2 Target

LH2 Acceptance: x_F Bin 10



LD2 Target

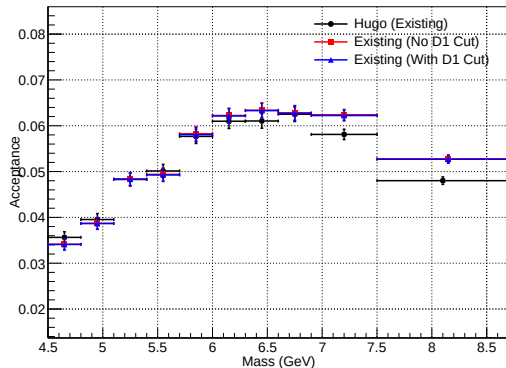
LD2 Acceptance: x_F Bin 10



Acceptance Comparison: x_F Bin 11

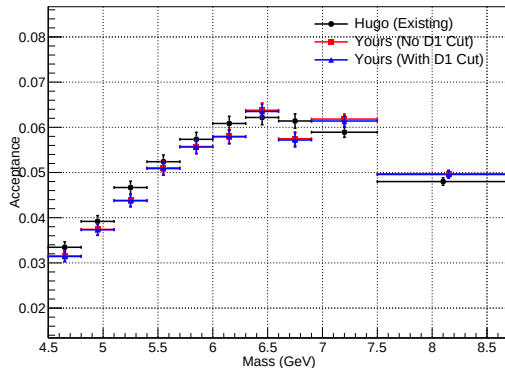
LH2 Target

LH2 Acceptance: x_F Bin 11



LD2 Target

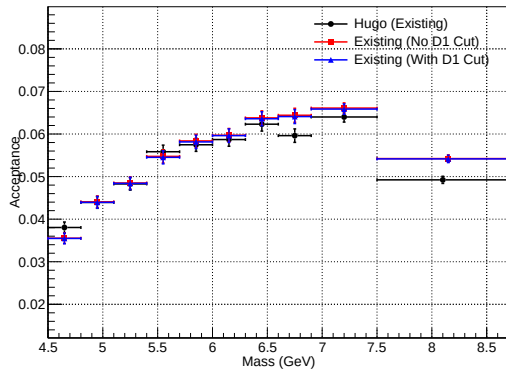
LD2 Acceptance: x_F Bin 11



Acceptance Comparison: x_F Bin 12

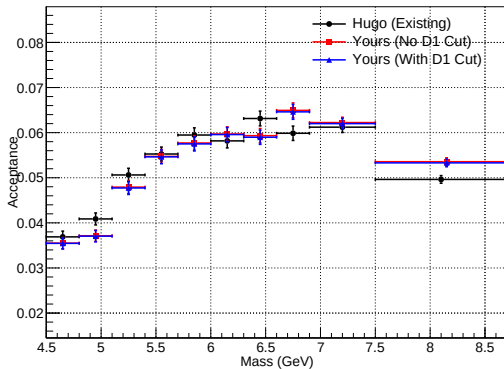
LH2 Target

LH2 Acceptance: x_F Bin 12



LD2 Target

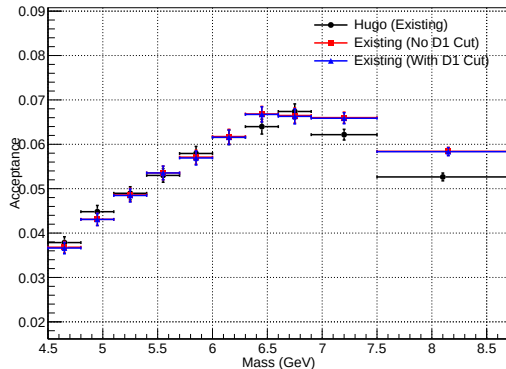
LD2 Acceptance: x_F Bin 12



Acceptance Comparison: x_F Bin 13

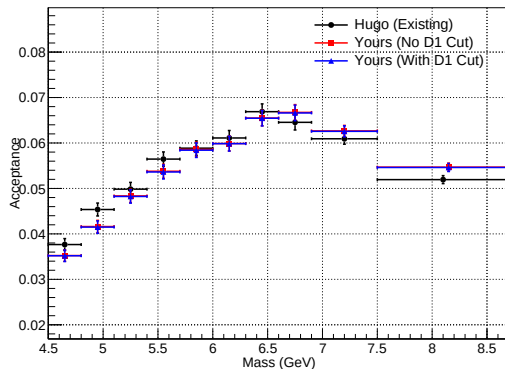
LH2 Target

LH2 Acceptance: x_F Bin 13



LD2 Target

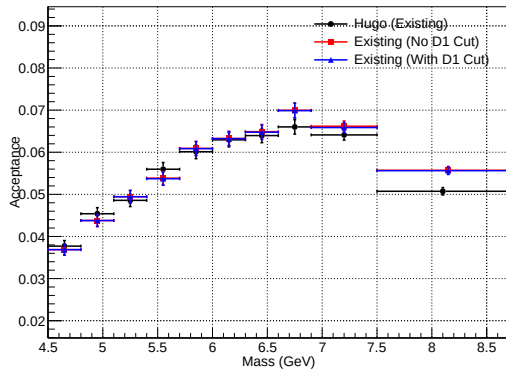
LD2 Acceptance: x_F Bin 13



Acceptance Comparison: x_F Bin 14

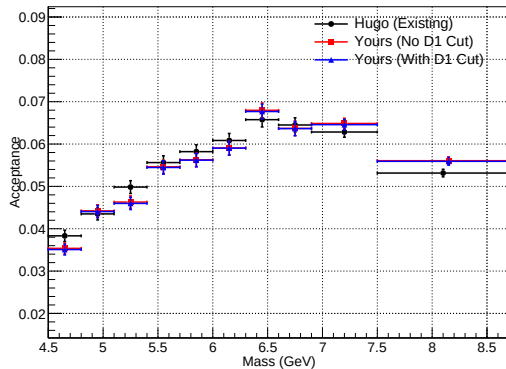
LH2 Target

LH2 Acceptance: x_F Bin 14



LD2 Target

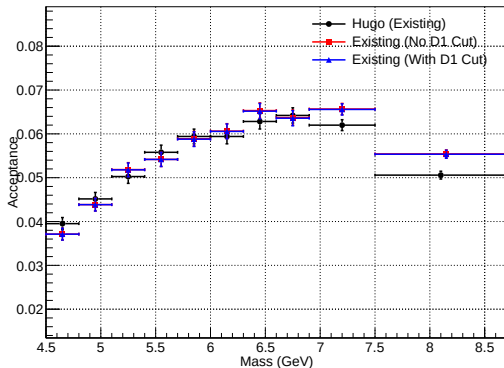
LD2 Acceptance: x_F Bin 14



Acceptance Comparison: x_F Bin 15

LH2 Target

LH2 Acceptance: x_F Bin 15



LD2 Target

LD2 Acceptance: x_F Bin 15

